Beyond the Average Man: The Art of Unlikelihood

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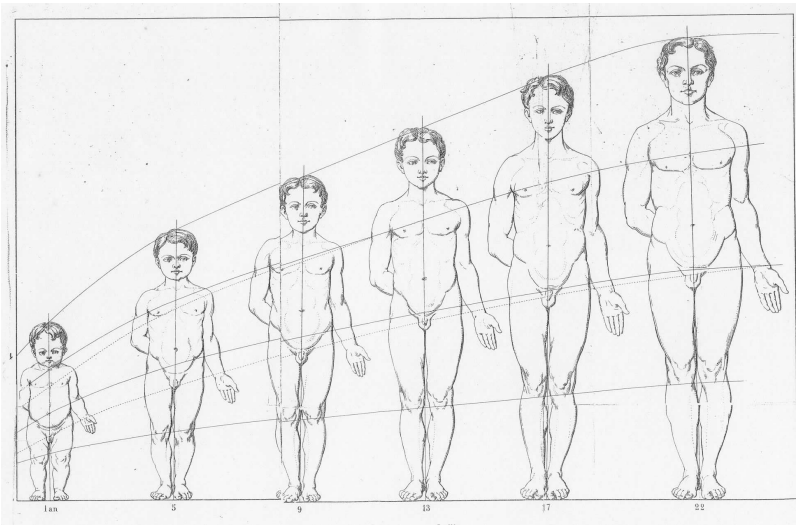

An Outline

- A Brief Biography of the Average Man
  - Science Begins with Measurement (Kelvin)
- Convergence versus Diversity
  - Heterogeneity and the Regression Fallacy
- Skepticism and Unlikelihoods
  - Robustness: Local versus Global Models
  - L_1 Regularization as Occam's (New) Razor
- Progress in (Quantile) Regression: Four Directions
  - Additive Non-parametric Models
  - Longitudinal (Panel) Data
  - Nonlinear Filtering
  - Multivariate Conditional Quantiles

The Father of the Average Man

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Through systematic measurement and relentless averaging Quetelet sought to extract man's essential qualities: social, economic, aesthetic, and moral.

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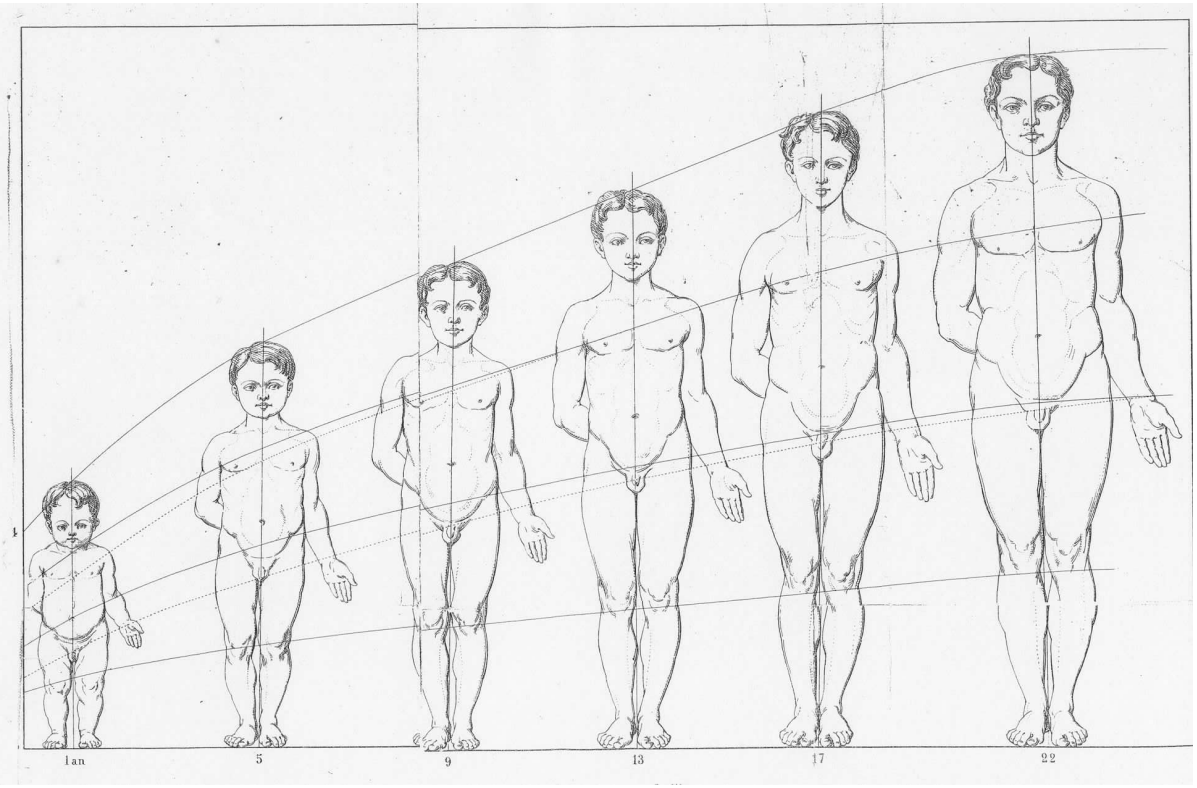
To Nightingale every piece of legislation was an experiment in the laboratory of society deserving study and demanding evaluation.

Portrait of the Average Man



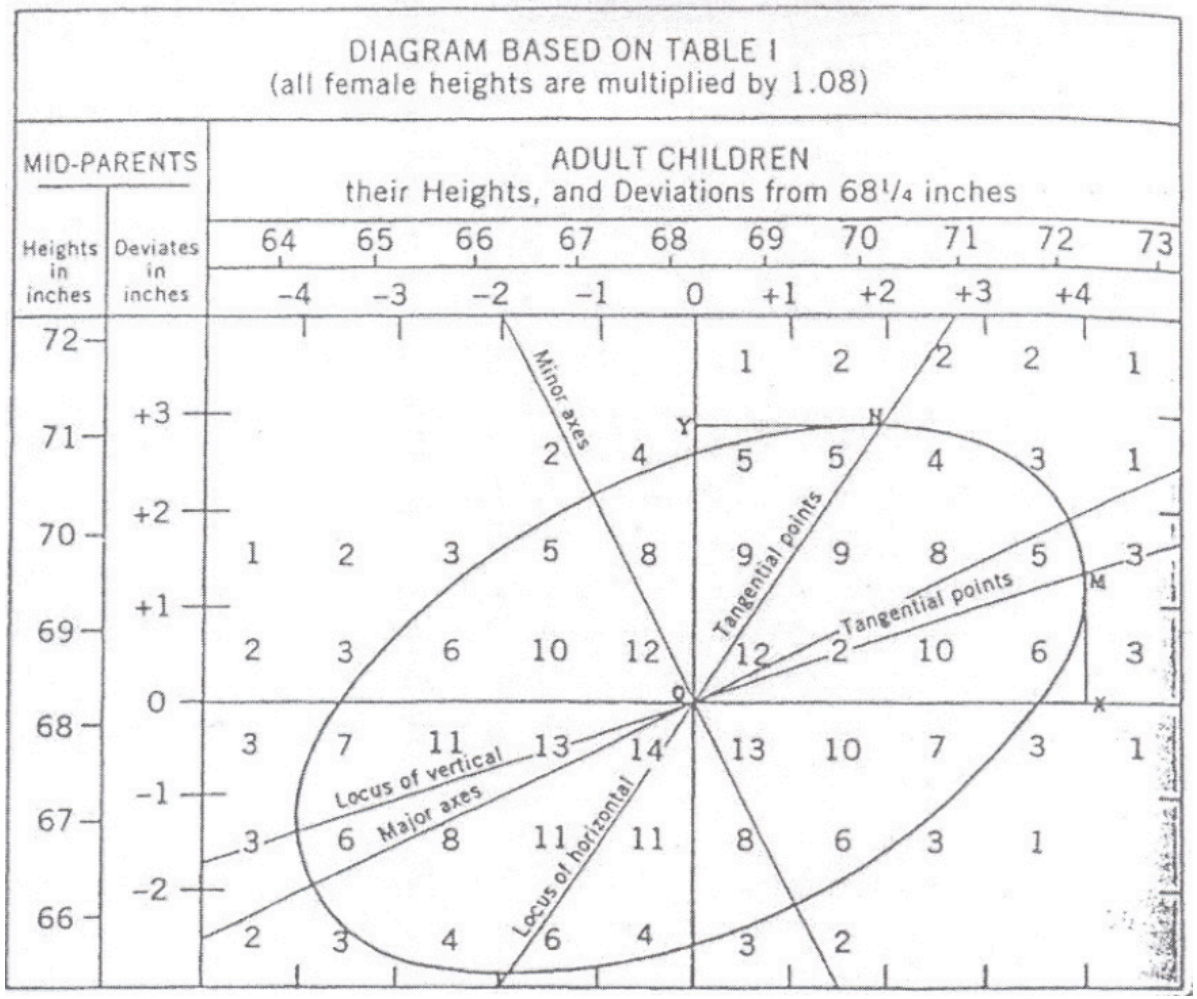
Quetelet's (1871) Anthropométrie

The Schoolmaster of the Average Man

Francis Galton (1822 - 1911)

Victorian adventurer, polymath and inveterate data collector and analyst. Galton's (1885) discovery of regression and correlation paved the way for the Average Man as a scientific construct.

Galton's Regression to Mediocrity



Galton's Discovery of Regression (1885)

Galton's "Most Likely" Children

Children are more mediocre than their parents:

- Children of midparents 3 inches taller than average are most likely to be 2 inches taller than average.
- Children of midparents 3 inches shorter than average are most likely to be 2 inches shorter than average.

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But parents are also more mediocre than their children:

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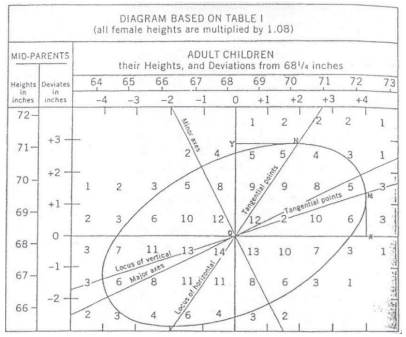
Galton's Bivariate Gaussian Sandbox



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- Conditional mean heights are linear in midparents' height.

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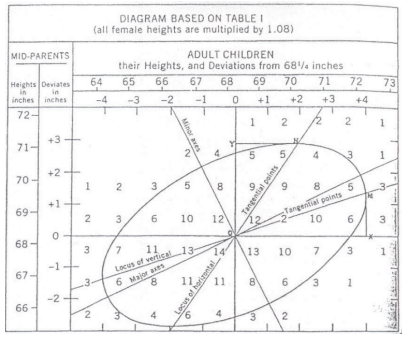

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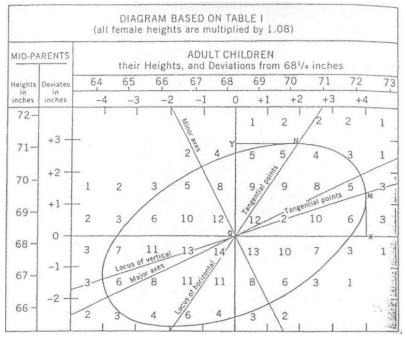

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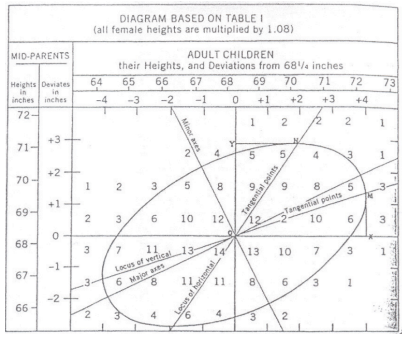

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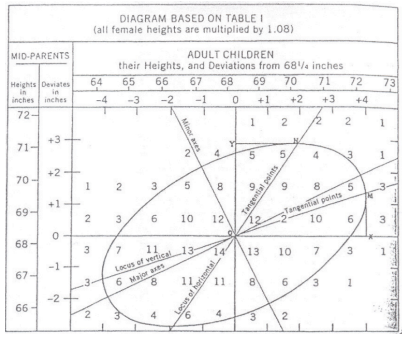

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- Means, medians and modes (most likely heights) coalesce.
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It all seems a little "too good to be true" – and even for heights it is, at best, only a decent first approximation.

Convergence versus Diversity

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Quetelet already exhibits symptoms of this malady:

> *"The more knowledge is diffused, so much the more do the deviations from the average disappear; and the more, consequently, do we tend to approach that which is beautiful, that which is good." [On Man, p.108]*

Secrist and the Regression Fallacy

Stigler's (1996) paper "The History of Statistics in 1933" provides a classic economic case study:

- In 1933 Horace Secrist published *The Triumph of Mediocrity* that advanced the alarming thesis that American business was inexorably tending toward mediocrity: Groups of highly profitable firms, over time, on average lost profitability, less profitable firms gained.

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- Harold Hotelling wrote a devastating review in JASA explaining that this was simply Galton's regression effect: firms were not – nor were human heights – converging to their respective means.
- Milton Friedman, a student of Hotelling's in 1933, felt compelled to write an essay for the JEL in 1992 called "Will Old Fallacies Ever Die?" to combat similar foolishness in contemporary economics.

The Demise of the Average Man

Robert Doisneau's (1952) "[Representative] Consumer"

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We have converged – not the the ideal world of Quetelet's Average Man, but to the realization that we need to take heterogeneity more seriously.

The Representative Agent – An Inconvenient Fiction

Heterogeneity is not the inessential cloud obscuring the ideal average man, it is precisely this diversity that we need to better understand.

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- Aggregation does not preserve economic rationality, the average man is not *homo economicus*: Sonnenschein, Debreu ...
- Treatment effects are often heterogeneous, and this is crucial to policy evaluation: Heckman, ...

Normal Skepticism and Unlikelihood

Before his conversion to the Gaussian faith, Galton was a confirmed nonparamet-nik. His obsessive data analysis relied almost exclusively on:

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Sufficiency of the mean and standard deviation require a dangerous leap of Gaussian faith, rarely justified when samples are moderately large.

Data Analysis versus Analysis of Data

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This tension is, in effect, the interplay between Tukey's exploratory and confirmatory data analysis.

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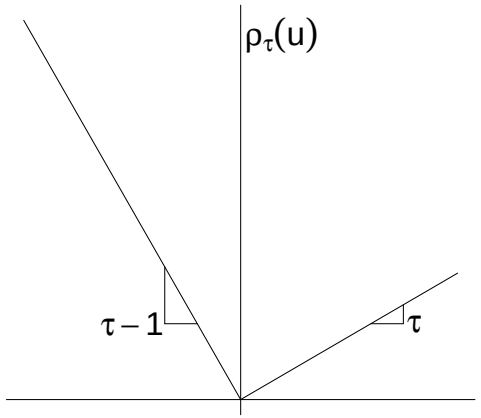

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$$\hat{\gamma}(\tau) = \text{argmin}_{\gamma} \sum \rho_\tau(y_i - g(x_i, \gamma)) \quad \tau\text{th nonlinear regression quantile}$$

Nonparametric Quantile Regression

One can easily do univariate locally polynomial QR by kernel weighting:

$$\min_{\gamma} \sum \rho_\tau(y_i - \gamma_0 - (x_i - x)\gamma_1 - \cdots - (x_i - x)^p \gamma_p) K_h(x - x_i),$$

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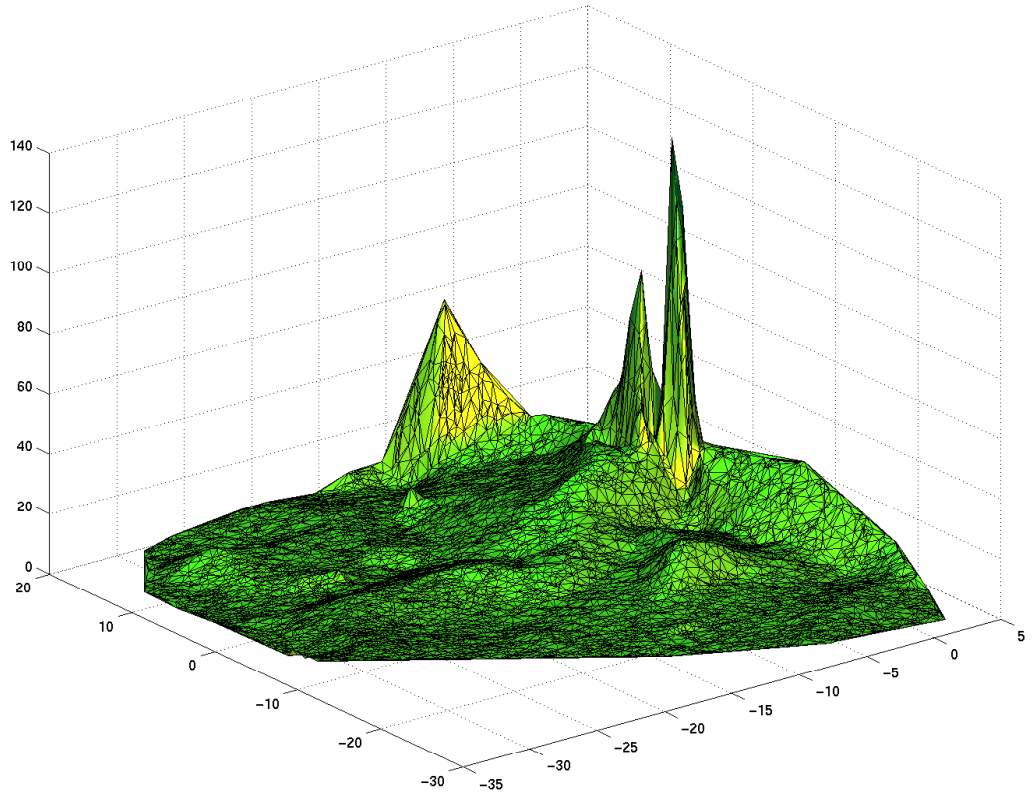
Where the penalty term can be interpreted as total variation of ∇g. Additive models of this type can also be easily estimated:

$$\min_{g} \sum_i \rho_\tau(y_i - \sum_j g_j(x_{ij})) + \sum_j \lambda_j \int \|\nabla^2 g_j\| dx.$$

In "R" such models can be estimated with `rqss` in my `quantreg` package.

Chicago Land Values via TV Regularization



Chicago Land Values: Based on 1194 vacant land sales and 7505 "virtual" sales introduced to increase the flexibility of the triangulation.

Quantile Regression for Panel Data

Time-series and panel data offer many interesting challenges, again regularization is a critical device. Consider the panel model,

$$Q_{Y_{it}}(\tau|x_{it}) = \alpha_i + x_{it}^\top \beta(\tau)$$

which can be estimated by solving,

$$\min \sum_{j=1}^{m} \sum_{i=1}^{n} \sum_{t=1}^{T} \rho_\tau(y_{it} - \alpha_i - x_{it}^\top \beta(\tau_j)) + \lambda \|D\alpha\|_1.$$

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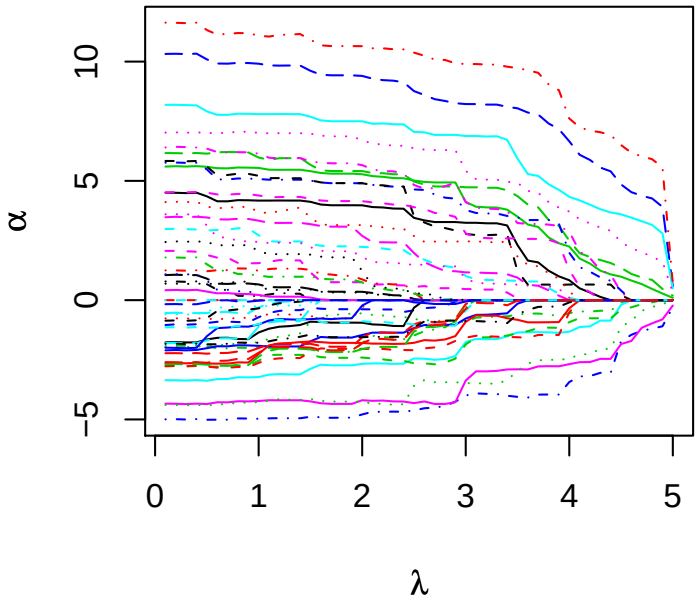
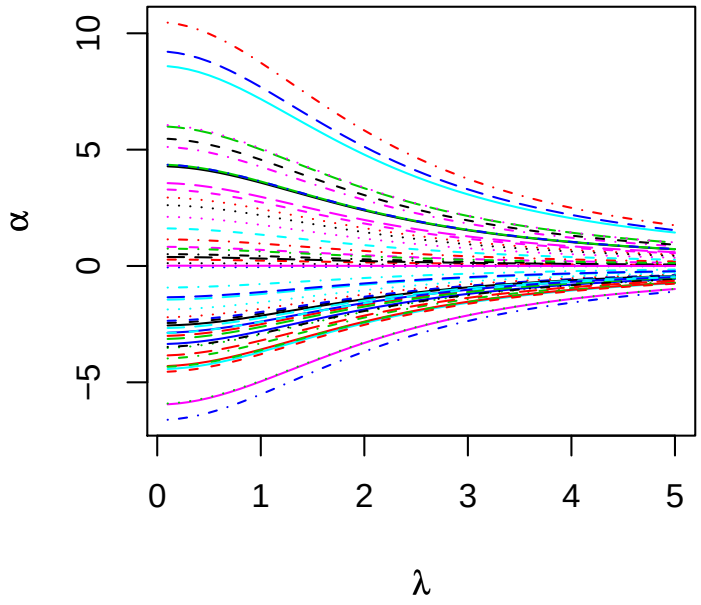
The regularization penalty shrinks the "fixed effects" toward a common value. Such problems are now feasible for large m, n, T thanks to recent developments in

- Barrier methods for linear programming, Frisch (1956), ...
- Sparse Linear Algebra

Shrinkage: ℓ_1 versus ℓ_2 Penalties





The ℓ_1 penalty tends to shrink most fixed effects to zero very quickly, while the ℓ_2 penalty imposes much more gradual shrinkage.

Regularization Penalties and Model Selection

Classical ℓ_2 regularization methods originating in the work of Tihkonov and Stein are too gentle; the polyhedral nature of ℓ_1 penalties are much better suited to model selection:

- Lasso/Lars of Tibshirani, Efron, ...
- Basis Pursuit of Donoho, ...
- Dantzig Selector of Candes and Tao.

ℓ_1 regularization is an emerging growth area in statistics; promising applications to imaging, signal processing and encryption. Strongly dependent on recent developments in convex optimization.

Whispering Down the Line

Problem: Transmit $x \in \mathbb{R}^n$ over a noisey channel.

Encoding: Send $y = Ax$ for $A \in \mathbb{R}^{m \times n}$, $m \gg n$, and receive either:

$$\tilde{y} = Ax + u \tag{1}$$

$$\tilde{y} = Ax + u + v \tag{2}$$

where $u \sim$ Gaussian, and v is (sparsely) arbitrarily bad.

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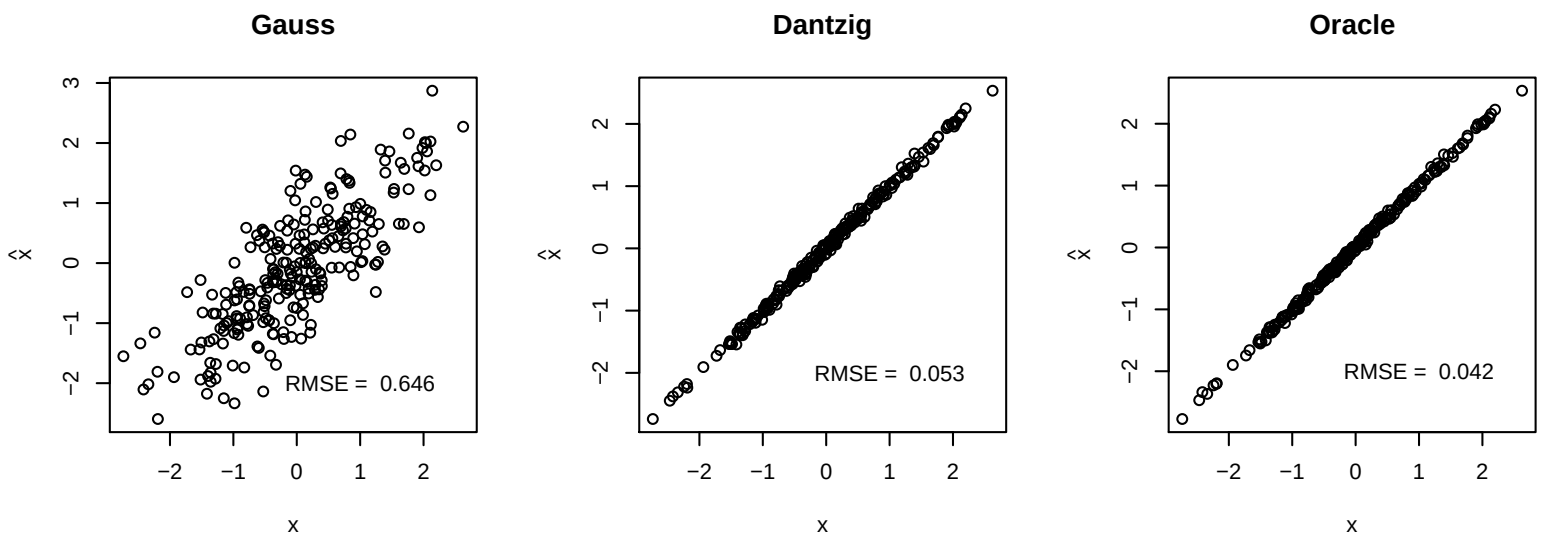
$$\hat{x} = (A^\top A)^{-1}A^\top (\tilde{y} - \tilde{v}) \tag{2}$$

where: $\tilde{v} = \text{argmin}\{\|v\|_1 \text{ such that } \|Q(\tilde{y} - v)\|_\infty \leqslant K\}$

Dantzig Decoding: An Example

Let $n = 256$, $m = 512$, and entries in x, u and A be iid standard Gaussian, and let ν be the mixture: $\nu_i = 0.9\delta_0 + 0.1\delta_{-2y_i}$.



Dantzig decoding (2) achieves almost the same accuracy as if ν were known.

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- Chesher's recent work on endogeneity in QR Models.

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Model: Let x denote a vector of covariates and H and W denote height and weight:

$$Q_H(\tau|x) = h(x; \gamma_H(\tau))$$

$$Q_W(\tau|H, x) = g(H, x; \gamma_W(\tau))$$

where, for convenience, g and h are linear in parameters, after expanding the covariates in B-spline basis functions.

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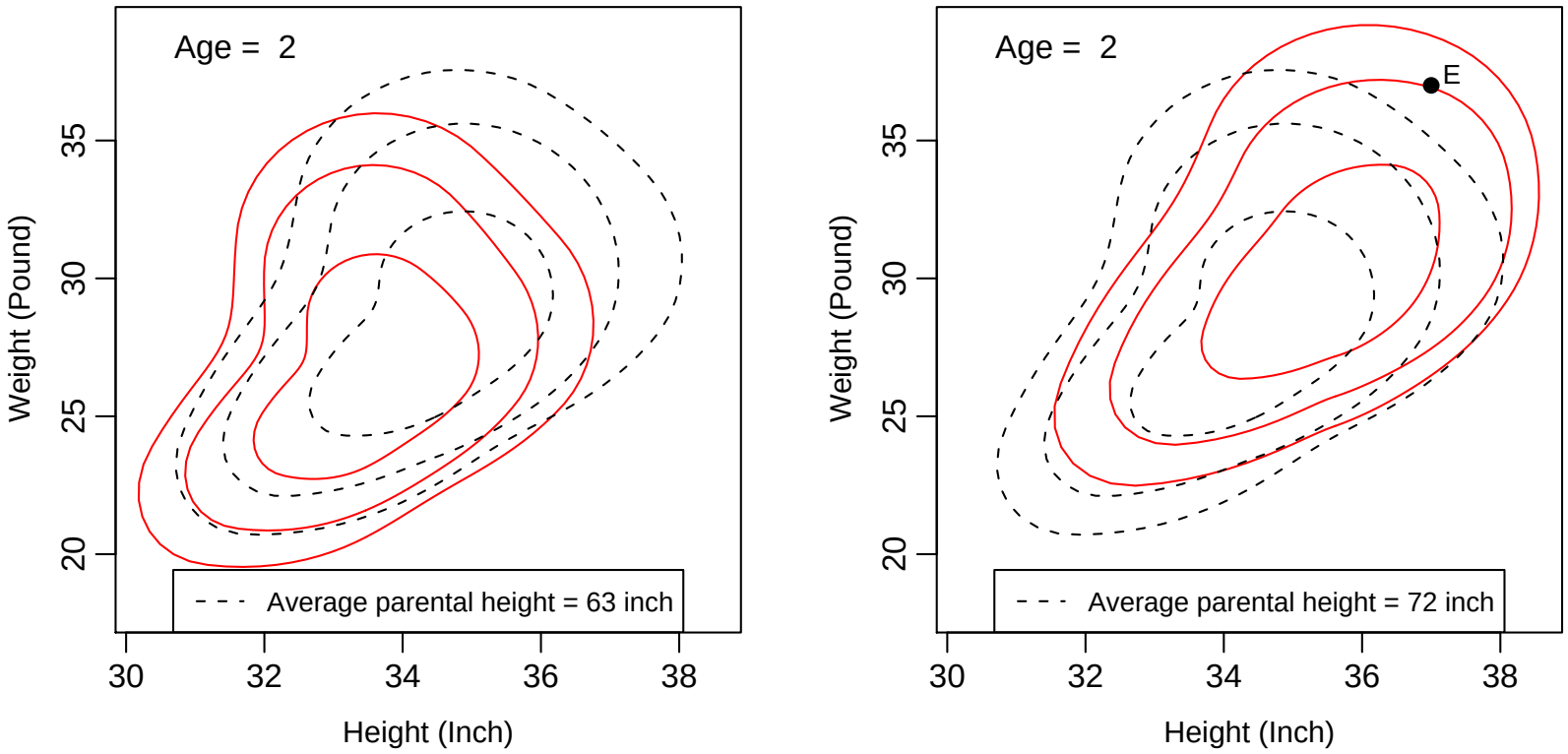
Simulation: Fix x, and draw pairs (τ_H, τ_W) from $(U_1, U_2) \sim U[0,1]^2$ and evaluate $(Q_H(U_H|x), Q_W(U_W|Q_H(U_H|x), x))$ and now estimate density contours, Rosenblatt (1952).

Height-Weight Contours for Two Two-Year Olds



Dashed lines are reference quantile contours for $\tau \in \{0.5, 0.8, 0.9\}$ for two year olds. Red contours conditional on parents height. Ying Wei (2007).

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Some progress has been made, but there are many challenging problems that lie ahead.